

Castalia

A four-channel electrochemical sensing SoC with a five-hart RISC-V back end

Maxx Seminario

University of Nebraska–Lincoln

September 20, 2026

Where cancers are caught, and whether anyone looks

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Target
application

The chip

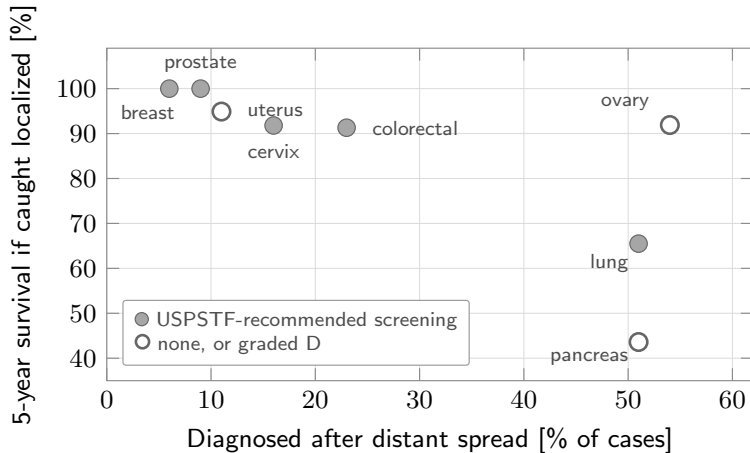
Analog front end

Against Mushkin
and the field

From currents to
a score

References

Backup



[1] SEER 21, 2016–2022; [2, 3] USPSTF

Why ovarian: late, lethal, and unscreened

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Target application

The chip

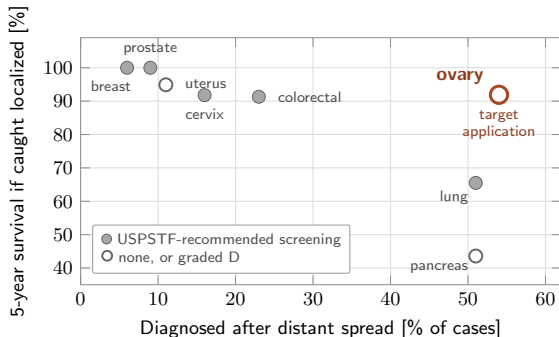
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Site	Distant at dx	Deaths/100 cases	Screening (USPSTF)
Ovary	54 %	59	D
Pancreas	51 %	78	D
Lung	51 %	54	B, LDCT
Colorectal	23 %	35	A
Cervix	16 %	31	A
Uterus	11 %	21	none
Prostate	9 %	11	C, PSA
Breast	6 %	13	B

USPSTF: A/B recommended; C offered selectively; D recommended *against*; none: no statement [7].

[1] SEER, 2026 estimates; [2, 3] USPSTF

Most lethal gynecologic cancer [1]: 92 % survivable if caught localized, yet 54 % present after spread, and screening is graded D [2]. UKCTOCS moved stage, not mortality [4].

One serum aliquot, four sites, one score

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Target application

The chip

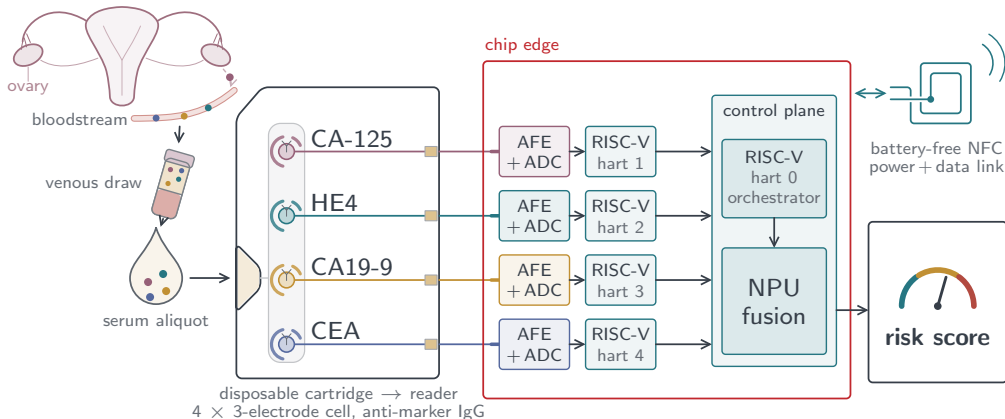
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Symptomatic-patient triage on a disposable four-site cartridge; four markers read at the same instant, fused on chip.

No single marker is a test; four fused are [5]

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The chip

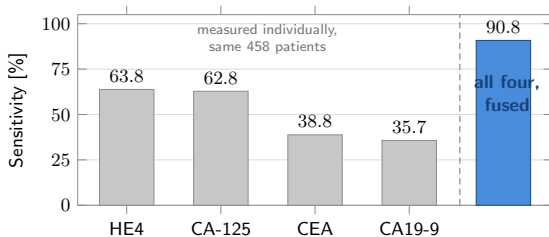
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Marker	Molecule	Cut-off
CA-125	MUC16 mucin epitope [8]	35 U/mL
HE4	WFDC2 glycoprotein [6]	70 pmol/L
CA19-9	sialyl-Lewis ^a antigen	37 U/mL
CEA	CEACAM5 glycoprotein	5 ng/mL

Each marker covers a subtype the others miss. The decision is a learned combination, so the chip is four channels plus the processor that fuses them.

Four markers resolved in one differential-pulse sweep

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Target application

The chip

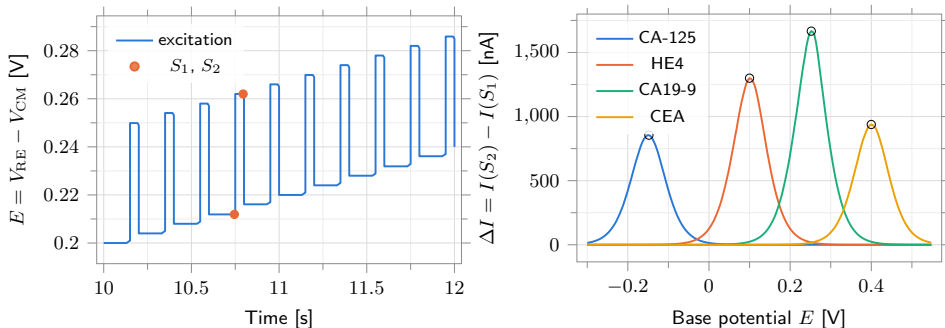
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■ schematic sim, pre-layout

Peak position identifies the reporter, peak height carries concentration. Apexes within 2 mV and 1 % of the assay targets; blank below 1 pA.

Every clinical cut-off lands inside the channel window

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The chip

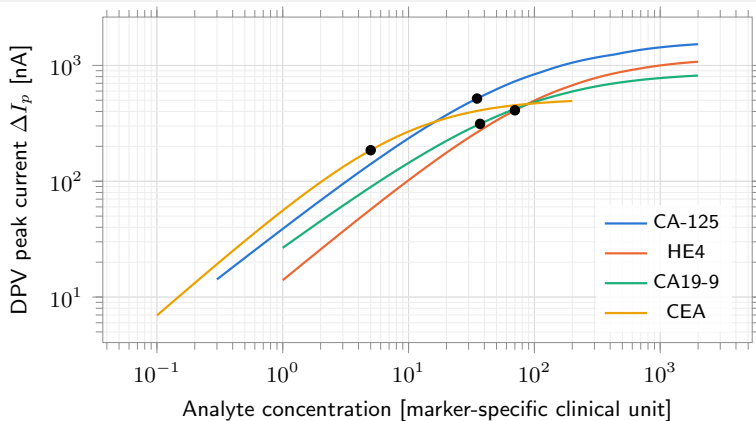
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■ schematic sim,

pre-layout

CA-125 35 U/mL, HE4 70 pmol/L, CA19-9 37 U/mL, CEA 5 ng/mL: all four on the steep part of the curve at one R_f setting.

Castalia at a glance

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From current to
future

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Backup

Digital

Harts	5 (1 orchestrator, 4 channel tiles)
ISA	rv32imac_zba_zbb_zbs_zbc
Clock	24 MHz rated, closed at 25 MHz
Measured CPI	1.260 aggregate, nine benchmarks
Private TCM	8 KiB per hart, 1-cycle
Shared RAM	80 KiB behind a round-robin arbiter
Synchronisation	16 hardware mutexes, LR/SC, AMOs
Interrupts	claim/complete, any vector to any hart
Accelerator	NPU: MLP, conv-1D, XNOR, GEMM
Power domains	4 MTCMOS, one per channel tile
Debug	JTAG, RISC-V Debug Module
Field boot	NFC ISO 14443A harvested boot

Analog and physical

Measurement sites	4 identical, simultaneous
Topology	bipolar grounded-WE potentiostat
Electrode pads	12 (CE/RE/WE $\times$ 4)
Converter	10-bit SAR per channel, extracted
Reference DAC	12-bit R-2R per channel
Feedback resistor	$18\text{ k}\Omega + 6\text{ k}\Omega \cdot N$, 320 codes
Full scale	$\pm 40.7\text{ }\mu\text{A}$ to $\pm 0.41\text{ }\mu\text{A}$
EIS	firmware mode of the same channels
Analog supply	2.5 V , $v_{\text{cm}} = 1.25\text{ V}$
Process	65 nm CMOS, 1P8M, HVT cells
Die / package	$3.0 \times 3.0\text{ mm}$ / LQFP-100
Digital power	10.72 mW post-route, SS, 25 MHz

Digital numbers from the hardened flow; analog numbers schematic-level, except the SAR, which is extracted.

Block diagram

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Target application

The chip

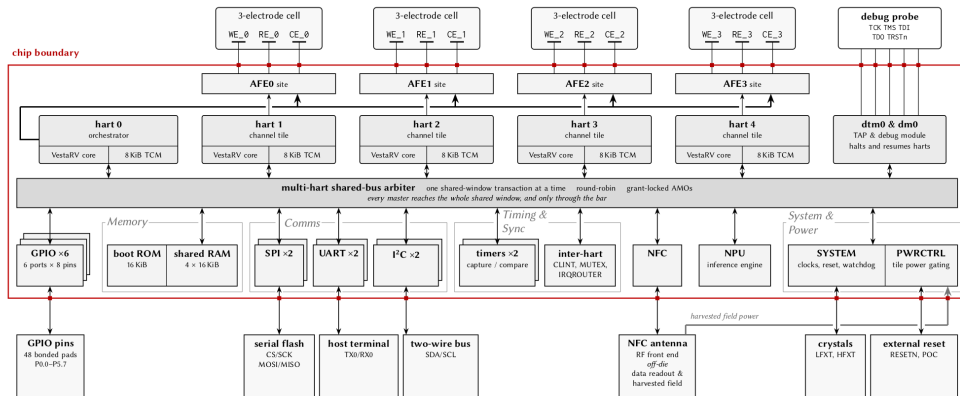
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Four channel tiles own four AFE sites; every master reaches ROM, RAM, NPU and peripherals only through the arbiter.

Physical implementation

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**ARTWORK
PENDING**



chip top, 3.0×3.0 mm
(Virtuoso export)

**ARTWORK
PENDING**



one channel
tile (Virtuoso
export)

Process	65 nm, 1P8M, HVT
Core box	$2690 \times 2690 \mu\text{m}$
Tile	$660 \times 880 \mu\text{m}$
Tile logic	≈ 98 k NAND2-eq.
Tile SRAM	8 KiB
Package	LQFP-100, 14×14 mm

■ hardened flow, post-route

One tile hardened once, instantiated four times; the notch in each tile is the analog channel's keep-out.

Digital, as closed

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Whole chip, post-route

Internal	3.78 mW
Switching	2.33 mW
Leakage	4.61 mW
Total	10.72 mW

One channel tile

Total	1.761 mW
Gated	0

Measured CPI, timed kernels

Benchmark	Instr.	CPI
spmv	802 447	1.143
multiply	20 892	1.163
vvadd	2 408	1.374
median	4 015	1.398
qsort	123 205	1.398
rsort	146 548	1.412
dhystone	200 523	1.504
towers	3 747	1.682
memcpy	11 017	1.726
Aggregate	1 314 802	1.260

Timing at 25 MHz

reg2reg WNS	+99 ps
Leakage share	43 %

■ hardened flow, post-route

SS 0.9 V 125 °C, statistical activity 0.2. Gating one tile removes 1.76 mW; the always-on band pays nothing.

Power gating: one switched rail per channel tile

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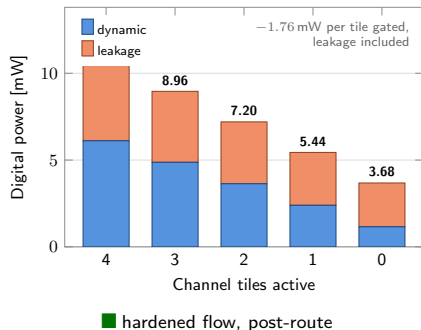
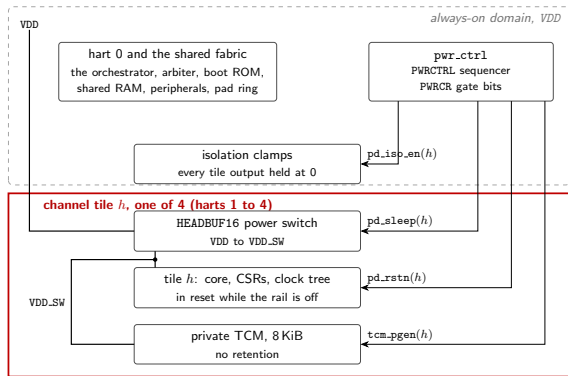
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Cold gating: the header cuts the rail, so a parked tile draws neither dynamic nor leakage power. Each tile gated removes 1.76 mW; the always-on band keeps 3.68 mW.

Grounded-WE two-amplifier potentiostat

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The chip

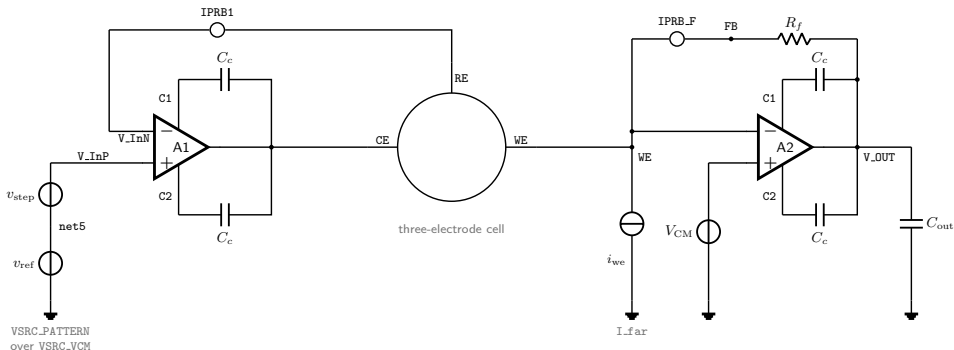
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A1 drives CE with RE in feedback; A2 holds WE at v_{cm} through R_f . $v_{out} = v_{cm} - I_{we}R_f$, both polarities.

Analog top level

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ARTWORK PENDING



Virtuoso export of the analog top level: four channels, shared bias, test bus

The class-AB amplifier: one cell, both roles

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Target application

The chip

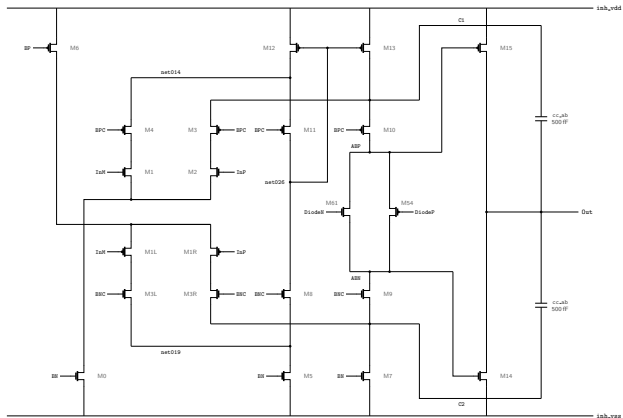
Analog front end

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From currents to a score

References

Backup



complementary
input pairs and
their cascodes

diode-connected
summing branch

class-AB
drive

output

testbench
compensation

The amplifier across corners

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The chip

Analog front end

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Backup

Parameter	Unit	tt	ff	fs	sf	ss	Spread
Open-loop gain A_{OL}	dB	82.77	79.88	83.14	83.37	84.94	5.06
Phase margin	°	73.52	74.70	74.04	72.86	73.34	1.84
Dominant pole $f_{-3\text{dB}}$	kHz	1.592	2.286	1.538	1.494	1.225	1.061
Gain-bandwidth product	MHz	22.781	23.187	22.894	22.939	22.587	0.600
Parameter	Unit	tt	ff	fs	sf	ss	Spread
Input offset voltage	mV	0.622	0.649	0.624	0.569	0.456	0.193

■ schematic sim, pre-layout

The amplifier under mismatch

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Target application

The chip

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Against Moshkin and the field

From currents to a score

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Backup

Parameter	Unit	N	Mean	σ	p_1	p_{99}	Worst	Spec	Yield	4 ch
Open-loop gain A_{OL}	dB	200	82.16	1.06	80.04	84.60	79.50	≥ 40.00	100.0	100.0
Phase margin	$^\circ$	200	73.62	1.42	69.80	76.21	68.93	≥ 40.00	100.0	100.0
Dominant pole $f_{-3\text{dB}}$	kHz	200	1.723	0.276	1.158	2.453	–	–	–	–
Gain-bandwidth product	MHz	200	22.815	3.566	16.191	31.741	–	–	–	–

Parameter	Unit	N	Mean	σ	Mean $\pm 3\sigma$	Min	Max	Spec	Yield	4 ch
Input offset voltage	mV	200	0.583	5.201	0.583 ± 15.604	–14.130	15.709	± 1.000	12.0	0.0

■ Monte Carlo, 200 samples

Bipolar accuracy through zero

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Target application

The chip

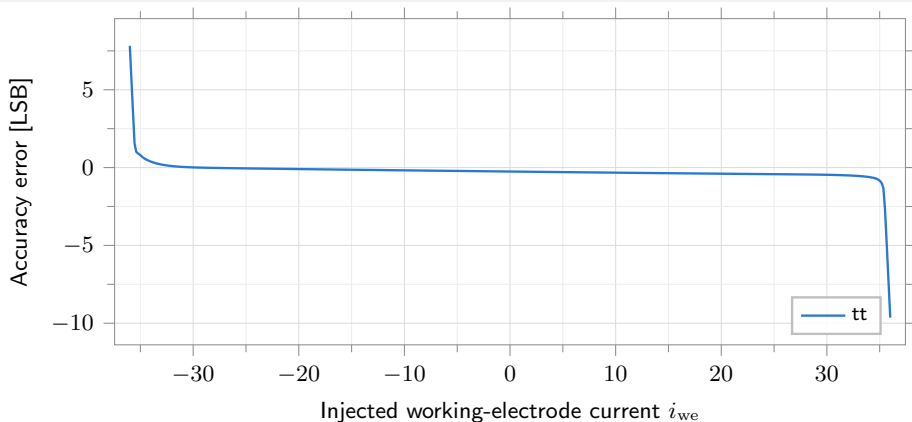
Analog front end

Against Moshkin and the field

From currents to a score

References

Backup



■ schematic sim, pre-layout

No kink at the crossing: class-AB output, no dead zone. This is the measurement rev-1's PMOS mirror could not make.

Cyclic voltammetry through the assembled channel

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Target application

The chip

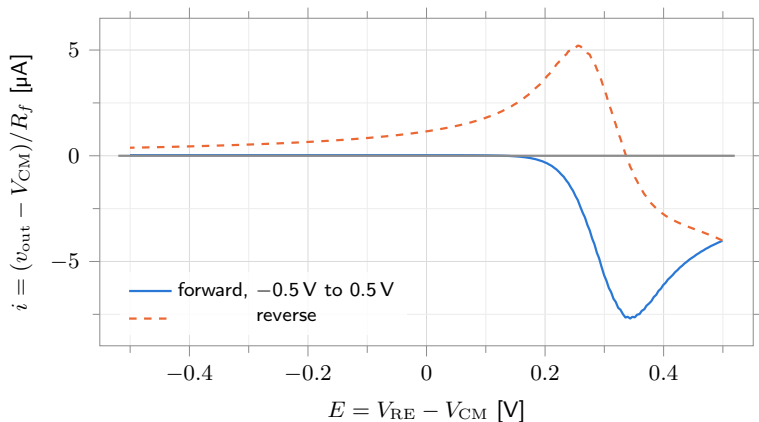
Analog front end

Against Mishkin and the field

From currents to a score

References

Backup



■ schematic sim,

pre-layout

Staircase potential from the on-chip DAC, current digitised by the on-chip SAR, unclipped over the sweep.

Programmable feedback resistor: one string, 320 codes

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Target application

The chip

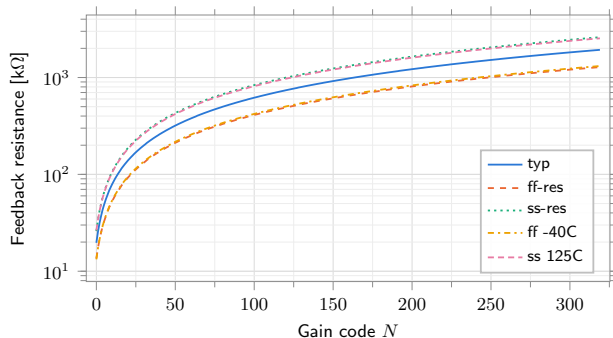
Analog front end

Against Mushkin and the field

From currents to a score

References

Backup



$$R_f = 18 \text{ k}\Omega + 6 \text{ k}\Omega \cdot N$$

N	R_f	Full scale
0	19.6 kΩ	±40.7 μA
63	397 kΩ	±2.02 μA
319	1.93 MΩ	±0.41 μA
LSB at 319		0.824 nA

Strictly monotonic at every corner, including the four thermometer seams. Maximum gain is noise-limited, not quantisation-limited.

Feedback resistor across corners and under mismatch

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Target application

The chip

Analog front end

Against Moshkin and the field

From currents to a score

References

Backup

Parameter	Unit	typ
Chord resistance	k Ω	19.639
Worst deviation	mV	1.998
Worst deviation	ADC LSB	1.255
Secant- R modulation	%	0.504

Parameter	Unit	N	Mean	σ	Yield
$R(N=0)$	k Ω	200	19.650	0.071	100.0
$R(N=63)$	k Ω	200	396.4515	0.3608	100.0
$R(N=64)$	k Ω	200	403.8759	0.3644	100.0
$R(N=319)$	M Ω	200	1.93212	0.00071	100.0
Seam step 63 $\rightarrow$ 64	k Ω	200	7.424	0.504	100.0

■ schematic sim, pre-layout

■ Monte Carlo, 200 samples

10-bit SAR ADC, extracted netlist

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Target application

The chip

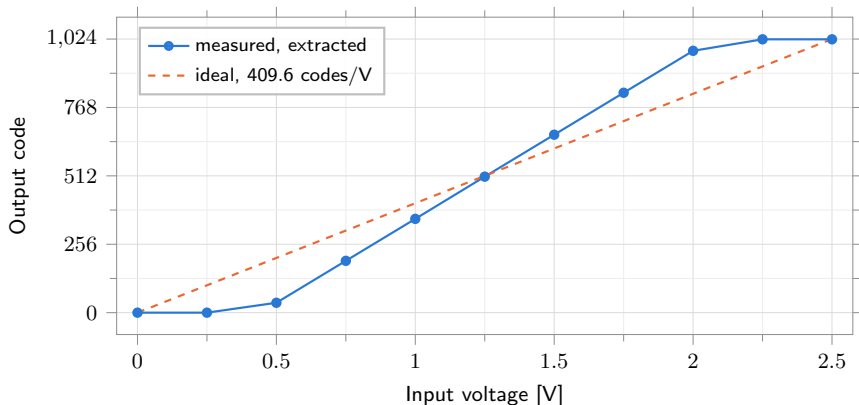
Analog front end

Against Moshkin and the field

From currents to a score

References

Backup



■ extracted

netlist

23 clocks per conversion, 870 kSa/s, 50 pJ. The capacitor array exists only in layout, so this is the one block characterised extracted.

SAR ADC across corners

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Target application

The chip

Analog front end

Against Moskin and the field

From currents to scores

References

Backup

Parameter	Unit	tt 37 °C	ff	fs	sf	ss
Fitted slope	codes/V	628.59	628.59	628.40	628.73	628.13
Effective LSB	mV	1.5909	1.5909	1.5914	1.5905	1.5920
Input at code 0	V	0.4408	0.4408	0.4407	0.4408	0.4399
Usable span	V	1.6274	1.6274	1.6280	1.6271	1.6286
Worst residual	LSB	0.495	0.495	0.490	0.477	0.484
RMS residual	LSB	0.294	0.294	0.277	0.288	0.302
Monotonic points		20/20	20/20	20/20	20/20	20/20
Conversion period	μs	1.150	1.150	1.150	1.150	1.150
Analog supply current	μA	6.360	4.787	5.453	8.437	15.429
Total power	μW	18.09	16.77	15.77	23.74	43.06
Energy per conversion	pJ	20.8	19.3	18.1	27.3	49.5

■ extracted netlist

Reference DAC across corners and under mismatch

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Target application

The chip

Analog front end

Against Moskin and the field

From currents to a score

References

Backup

Parameter	Unit	tt	ff	fs	sf	ss	Spread
Zero-scale output	μV	0.02	0.25	0.01	0.06	0.00	0.25
Full-scale output	V	2.499388	2.499388	2.499388	2.499388	2.499388	0.000000
LSB size	μV	610.351	610.351	610.351	610.351	610.351	0.000
INL, most positive	LSB	0.113	0.101	0.100	0.127	0.130	0.030
INL, most negative	LSB	-0.376	-0.338	-0.390	-0.362	-0.414	0.076
INL, worst magnitude	LSB	0.376	0.338	0.390	0.362	0.414	0.076
DNL, most positive	LSB	0.002	0.001	0.002	0.002	0.002	0.000
DNL, most negative	LSB	-0.423	-0.374	-0.424	-0.424	-0.479	0.105
DNL, worst magnitude	LSB	0.423	0.374	0.424	0.424	0.479	0.105

Parameter	Unit	N	Mean	σ	Median	p_1	p_{99}	Worst	Spec	Yield	4 ch
INL worst	LSB	200	1.485	0.620	1.364	0.426	3.287	3.393	≤ 1.000	20.5	0.2
INL max	LSB	200	1.154	0.610	1.031	0.194	2.886	-	-	-	-
INL min	LSB	200	-1.485	0.620	-1.364	-3.287	-0.426	-	-	-	-
DNL min	LSB	200	-1.682	1.415	-1.287	-6.161	-0.058	-6.499	≥ -1.000	42.0	3.1
DNL max	LSB	200	1.123	1.128	0.713	0.007	4.760	-	-	-	-
DNL worst	LSB	200	2.249	1.314	1.847	0.441	6.161	6.499	≤ 0.750	9.0	0.0

■ schematic sim, pre-layout ■ Monte Carlo, 200 samples

How impedance is measured: the same loop, no extra hardware

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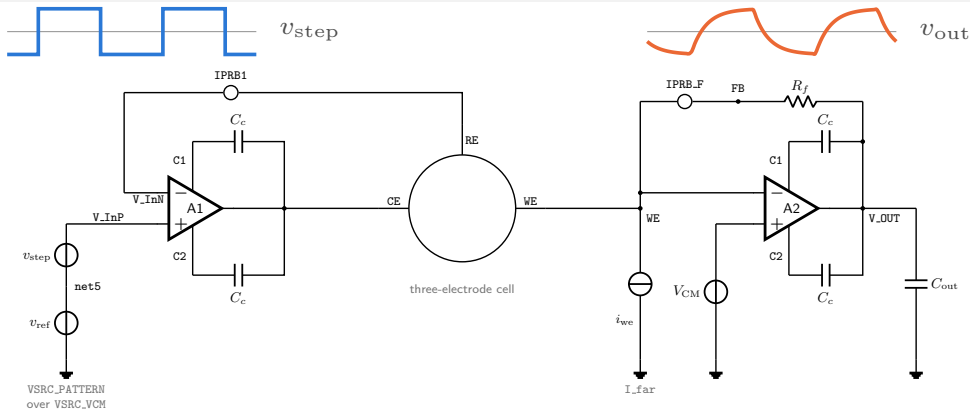
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Excite v_{step} toggles between two DAC codes, ± 4 or ± 10 mV about the DC setpoint v_{ref}
 Sense v_P at RE; $i_{WE} = -v_{out}/R_f$ through the same R_f a DC measurement uses
 Digitise one SAR, the two records interleaved; firmware I/Q multiply-accumulate over M periods
 Solve $\hat{Z} = -R_f \hat{V}_P / \hat{V}_{OUT}$, or a ratio to a known Z_{cal} , where R_f , the LSB and the square's $4/\pi$ cancel
The mode adds a clock divider and an accumulate to the channel, nothing else.

Impedance spectroscopy on the same channel

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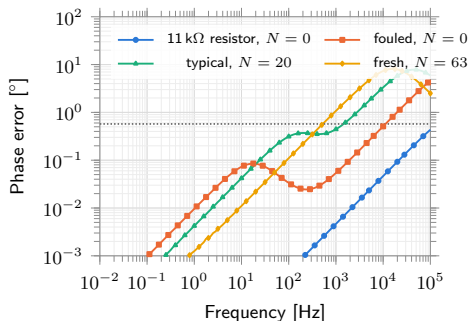
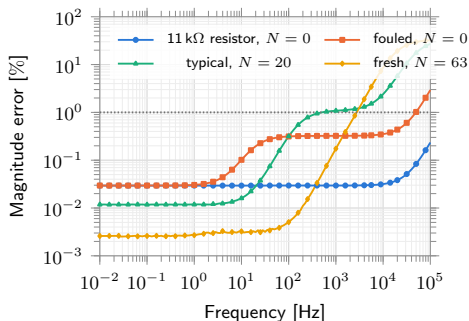
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Excitation	Demodulation	Accuracy	$ Z $ range
2-code DAC square	firmware, channel hart	$\pm 1\%$, $\pm 0.6^\circ$ to 5.05 kHz; $\pm 5\%$ to 26.5 kHz	246 Ω – 12.1 M Ω

■ schematic sim, pre-layout

No EIS engine. Impedance is a ratio of two readings through one converter; the DAC drops out of the accuracy budget.

One AFE channel, as simulated

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From currents to voltage

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Backup

Parameter	Value	Condition
CE compliance	13.2 mV – 2.480 V	2.5 V rail, corners
Cell current delivered	$\pm 52.5 \mu\text{A}$	12 k Ω cell
TIA compliance	$\pm 0.80 \text{ V}$ about v_{cm}	
Feedback resistor	19.6 k Ω – 1.93 M Ω , 320 codes	monotonic at every corner
Full-scale current	$\pm 40.7 \mu\text{A}$ ($N=0$) to $\pm 0.41 \mu\text{A}$ ($N=319$)	SAR window $\cap$ TIA compliance
Current quantum	0.824 nA at maximum gain	below the 2 nA noise budget
Bipolar accuracy	0.25–0.38 LSB worst	560 k Ω , 1 M Ω electrode, corners
Control-loop PM	79–80°	five corners, into a cell
TIA-loop PM	89–93°	12-cell electrode grid
Input current noise	341 pA standalone, 3.6–21 nA into an electrode	0.1–100 Hz, 560 k Ω
Zero-current offset	$\sigma(v_{\text{re}}) = 5.72 \text{ mV}$, gain-independent	200 MC samples, calibratable
Quiescent current	197–250 μA per channel	amplifiers + local bias

■ schematic sim, pre-layout ■ Monte Carlo, 200 samples

Pre-layout, electrode held nominal: no row includes electrode-to-electrode spread.

Against the field: the ISCAS'26 table, plus Castalia

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Parameter	Castalia (this work)	Myshkin [9]	Chen '19 [10]	Ghoreishizadeh '17 [11]	Lin '23 [12]
Modality	CA/CV/DPV/EIS	CA/CV	CA/CV	CA/CV	CV/EIS
Channels	4, simultaneous	1	1	1, differential	1
Current direction	bipolar	unipolar	bipolar	bipolar	bipolar
Sensitivity (pA)	341 ^a	56.5 ^b	41	0.47	1.7
$I_{\max}$ (μA)	±40.7	+20	±5	±20	-48/+16
Power (mW)	3.68 ^c	3.0 ^d	0.016	9.3	1.0
Process (nm)	65	65	180	350	180
On-chip processor	5 RISC-V harts	1 RISC-V hart	none	none	none
On-chip ML	NPU, 4 modes	NPU, 1 mode	none	none	none

Castalia column is best case and pre-silicon. ^aAFE only: input-referred rms, 0.1–100 Hz, maximum gain, ideal R_f , no electrode; the converter LSB is 0.82 nA. ^bAFE only; ADC quantisation limits the system to 153 pA. ^cDigital only: hart 0 awake, all four channel tiles gated, post-route SS 125 °C 25 MHz; one channel measuring adds 1.76 mW (tile) + 1.14 mW (AFE). ^dMeasured: 2.8 mW digital at 24 MHz + 258 μW AFE.

Still the only entries with a processor and an ML engine on the die; Castalia adds concurrency, both current polarities and EIS.

Myshkin to Castalia: what the digital side gained

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	Myshkin	Castalia		Myshkin	Castalia
Harts	1	5	Power domains	memories	+4 MTCMOS tiles
ISA	rv32imc	rv32imac + Zb*	Field power	–	NFC, harvested
Shared window	–	80 KiB, arbiter	NPU modes	1	4
Sync	–	mutexes, CLINT, AMOs	Peripherals	8 kinds	17 kinds
IRQ vectors	83	124, any-to-any	Measured CPI	–	1.260
Debug	–	JTAG, Debug Module	AFE sites	1, unipolar	4, bipolar, EIS
Traps	legacy	standard M-mode	Die	1.5 mm ²	9.0 mm ²

Same core, same process, same 24 MHz rating. Everything above the core is new, and all of it is generated from one JSON file.

The fusion runs on the NPU

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$$s = \sigma \left(\sum_{k=1}^4 w_k \hat{c}_k + b \right)$$

Input	peak current, peak potential, baseline $\times 4$
Mode	0, dense layer, one sigmoid
Output	one risk score
Weights	trained off-chip, staged by firmware

Per site

Hart	1, ownership enforced in hardware
Potential	own 12-bit DAC
Converter	own 10-bit SAR
Power	own MTCMOS domain

The logistic form of the cleared two-marker algorithm, extended to four inputs and moved from the central laboratory onto the electrode.

References (1/2)

Castalia

Maxx Seminario

Target application

The drug

Analys front end

Against Moshkin and the field

From currents to a score

References

Backup

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The chip

Analogue front end

Against Moshiin
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From currents to
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Backup: USPSTF screening grades

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The dip

Analog front end

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From currents to a score

References

Backup

Grade	Certainty of net benefit	Definition [7]	Cancer screening [2, 3]
A	high, substantial	Recommends the service. Offer or provide it.	cervix (2018), colorectal 50–75 (2021)
B	high moderate, or moderate	Recommends the service. Offer or provide it.	breast 40–74 (2024), lung LDCT 50–80 (2021), colorectal 45–49 (2021) prostate PSA 55–69 (2018)
C	at least moderate, small	Offer selectively, by professional judgment and patient preference.	
D	no net benefit, or harms outweigh	Recommends <i>against</i> the service. Discourage its use.	ovary (2018) , pancreas (2019)
I	cannot be determined	Evidence insufficient, poor, or conflicting.	—
none	—	No USPSTF statement exists for the condition.	uterus

Ovarian D is a verdict on a test that was tried, CA-125 plus transvaginal ultrasound [2, 4], not on the absence of one.